

Lexie Linear Algebra Homework

Chapter 6 - NumPy Array Broadcasting

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Purpose: learn how NumPy does arithmetic between arrays of different shapes when the shapes are compatible. This is one of the most useful ideas in numerical Python, especially for feature scaling and machine learning data preparation.

Part A - Straightforward NumPy Practice

Before every operation, print the shapes. The habit of checking shapes is one of the best ways to avoid NumPy confusion.

```
import numpy as np
```

```
A = np.array([
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
])
```

- Add 5 to every element of A.
- Multiply every element of A by 2.
- Create `row_adjustment = np.array([1, 2, 3])` and add it to A.
- Create `col_adjustment = np.array([[100], [200], [300]])` and add it to A.
- Print the shapes of A, `row_adjustment`, and `col_adjustment`.
- Explain why `row_adjustment` broadcasts across the columns.
- Explain why `col_adjustment` broadcasts down the rows.

Part B - Practical Problem: Standardizing Features

This is the practical broadcasting pattern used constantly in data science: subtract a column mean and divide by a column standard deviation.

```
x = np.array([
    [2, 75],
    [4, 82],
    [1, 60],
    [5, 90],
    [3, 78],
    [2, 65],
    [6, 92],
    [1, 58]
], dtype=float)
```

- Compute column means with `means = X.mean(axis=0)`.
- Compute column standard deviations with `stds = X.std(axis=0)`.
- Compute `X_scaled = (X - means) / stds`.
- Print `X_scaled.mean(axis=0)` and `X_scaled.std(axis=0)`. These should be close to `[0, 0]` and `[1, 1]`.

- Explain why `X - means` works even though `X.shape` is `(8, 2)` and `means.shape` is `(2,)`.

Challenge Question

```
bad = np.array([1, 2, 3])  
X + bad
```

Explain why this fails. Your answer should mention the trailing dimensions: `X` has 2 columns but `bad` has length 3.

Reflection Questions

- What does broadcasting save us from writing manually?
- Why is `(8, 2)` compatible with `(2,)` but not with `(3,)`?
- Why is standardization easier with broadcasting than with a loop?